ARJUNA JEE 2027

Lecture -13 | Mathematics

TRIGONOMETRIC FUNCTIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Range of Trigonometric Functions**

B **Question Practice**

C

D



H.W. Discussion

Critical Thinking Question (CTQ)



Prove that :

$$\frac{\cos 3x}{\sin 2x \sin 4x} + \frac{\cos 5x}{\sin 4x \sin 6x} + \frac{\cos 7x}{\sin 6x \sin 8x} + \frac{\cos 9x}{\sin 8x \sin 10x} = \frac{1}{2}(\operatorname{cosec} x)[\operatorname{cosec} 2x - \operatorname{cosec} 10x]$$

$$T_1 = \frac{2\cos 3x \sin x}{2\sin x \cdot \sin 2x \cdot \sin 4x} = \frac{1}{2}\operatorname{cosec} x \left\{\frac{\sin(4x)}{\sin 2x \cdot \sin 4x} - \frac{\sin(2x)}{\sin 2x \cdot \sin 4x}\right\} = \frac{1}{2}\operatorname{cosec} x \cdot \{\operatorname{cosec} 2x - \operatorname{cosec} 4x\}$$

$$T_2 = \frac{2\cos 5x \cdot \sin x}{2\sin x \sin 4x \cdot \sin 6x} = \frac{1}{2}\operatorname{cosec} x \left\{\frac{\sin 6x}{\sin 4x \sin 6x} - \frac{\sin(4x)}{\sin 4x \cdot \sin 6x}\right\} = \frac{1}{2}\operatorname{cosec} x \cdot \{\operatorname{cosec} 4x - \operatorname{cosec} 6x\}$$

$$\vdots$$

Question

Prove that : $\sum_{r=0}^{n} \frac{\sin(3^r\theta)}{\cos(3^{r+1}\theta)} = \frac{\tan(3^{n+1}\theta) - \tan\theta}{2}$

$$T_1 = \frac{2\sin(\theta)\cos\theta}{2\cos 3\theta \cdot \cos\theta} = \frac{\overset{3\theta-\theta}{\overset{\uparrow}{\sin 2\theta}}}{2\cos 3\theta \cdot \cos\theta} = \frac{1}{2}\left(\frac{\sin 3\theta \cos\theta}{\cos 3\theta \cos\theta} - \frac{\cos 3\theta \sin\theta}{\cos 3\theta \cos\theta}\right)$$

$$T_2 = \frac{2\sin 3\theta \cos 3\theta}{2\cos 9\theta \cdot \cos 3\theta} = \frac{\overset{9\theta-3\theta}{\overset{\uparrow}{\sin 6\theta}}}{2\cos 9\theta \cdot \cos 3\theta} \rightarrow = \frac{1}{2}(\tan 3\theta - \tan\theta)$$

$$= \frac{1}{2}(\tan 9\theta - \tan 3\theta)$$

$$\vdots$$

Question

80%

Compute the value of the sum $\sum_{r=1}^{n}\left(\frac{\tan(2^{r-1})}{\cos(2^r)}\right)$

A $\tan 2^n - 1$

B $\tan 2^n - \tan 1$

C $\tan 2^{n-1} - \tan 1$

D $\tan 2^n + \tan 1$

$r=1 \Rightarrow T_1 = \frac{\tan 1}{\cos 2} = \frac{\sin(2-1)}{\cos 2 \cdot \cos 1} = \tan 2 - \tan 1$

$r=2 \Rightarrow T_2 = \frac{\tan 2}{\cos 4} = \frac{\sin(4-2)}{\cos 4 \cdot \cos 2} = \tan 4 - \tan 2$

Critical Thinking Question

[Ans: A, D]

Given $f(n,\theta)=\tan\frac{\theta}{2}(1+\sec\theta)(1+\sec 2\theta)(1+\sec 2^2\theta)\ldots(1+\sec 2^n\theta)$ where $n\in\mathbb{N}$, then which of the following is correct?

tanθ.

A $f\left(2,\frac{\pi}{16}\right)=1$

B $f\left(2,\frac{\pi}{16}\right)=\sqrt{2}-1$

C $f\left(2,\frac{\pi}{32}\right)=1$

D $f\left(2,\frac{\pi}{32}\right)=\sqrt{2}-1$

$$\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\times\left(\frac{\cos\theta+1}{\cos\theta}\right)=\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\times\frac{2\cos^2\frac{\theta}{2}}{\cos\theta}=\frac{\sin\theta}{\cos\theta}=\tan\theta$$

$$\tan\theta\cdot(1+\sec 2\theta)=\tan 2\theta.$$

$$\frac{\sin\theta}{\cos\theta}\times\left(\frac{\cos 2\theta+1}{\cos 2\theta}\right)=\frac{\sin\theta}{\cos\theta}\times\frac{2\cos^2\theta}{\cos 2\theta}=\frac{\sin 2\theta}{\cos 2\theta}.$$

Critical Thinking Question

[Ans: A, D]

Given $f(n,\theta)=\tan\frac{\theta}{2}(1+\sec\theta)(1+\sec 2\theta)(1+\sec 2^2\theta)\ldots(1+\sec 2^n\theta)$ where $n\in\mathbb{N}$, then which of the following is correct?

A $f\left(2,\frac{\pi}{16}\right)=1$

B $f\left(2,\frac{\pi}{16}\right)=\sqrt{2}-1$

C $f\left(2,\frac{\pi}{32}\right)=1$

D $f\left(2,\frac{\pi}{32}\right)=\sqrt{2}-1$

$\tan\theta(1+\sec 2\theta)$

$\tan 2\theta\cdot(1+\sec 2^2\theta)$

$\tan(2^2\theta)$

$f(n,\theta)=\tan(2^n\theta)$

Critical Thinking Question

[Ans: A, D]

Given $f(n,\theta)=\tan\frac{\theta}{2}(1+\sec\theta)(1+\sec 2\theta)(1+\sec 2^2\theta)\ldots(1+\sec 2^n\theta)$ where $n\in\mathbb{N}$, then which of the following is correct?

A $f\left(2,\frac{\pi}{16}\right)=1$

B $f\left(2,\frac{\pi}{16}\right)=\sqrt{2}-1$

C $f\left(2,\frac{\pi}{32}\right)=1$

D $f\left(2,\frac{\pi}{32}\right)=\sqrt{2}-1$

$f(n,\theta)=\tan(2^n\theta)$

(A) $n=2,\ \theta=\frac{\pi}{16}$

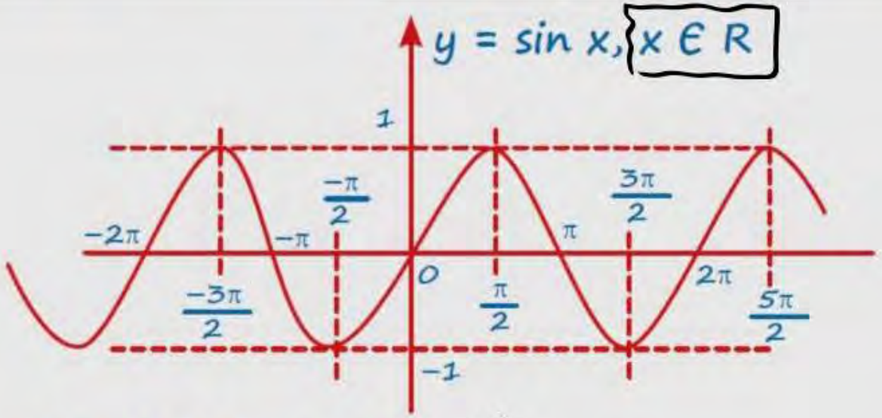
$f\left(2,\frac{\pi}{16}\right)=\tan\left(2^2\cdot\frac{\pi}{16}\right)=\tan\left(\frac{\pi}{4}\right)=1$

(C) $n=2,\ \theta=\frac{\pi}{32}$

$f\left(2,\frac{\pi}{32}\right)=\tan\left(2^2\cdot\frac{\pi}{32}\right)=\tan\left(\frac{\pi}{8}\right)=\sqrt{2}-1$

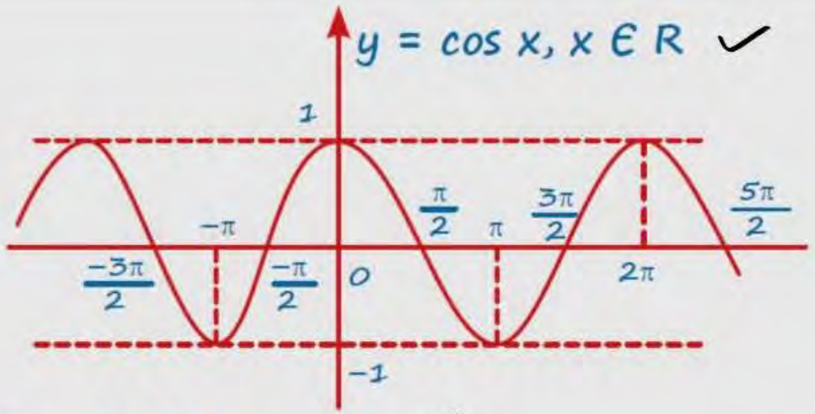
LAST CLASS RECAP

Graph of Trigonometric Functions: ✓



Range (sinx) ∈ [−1, 1] ✓

Period = 2π ✓



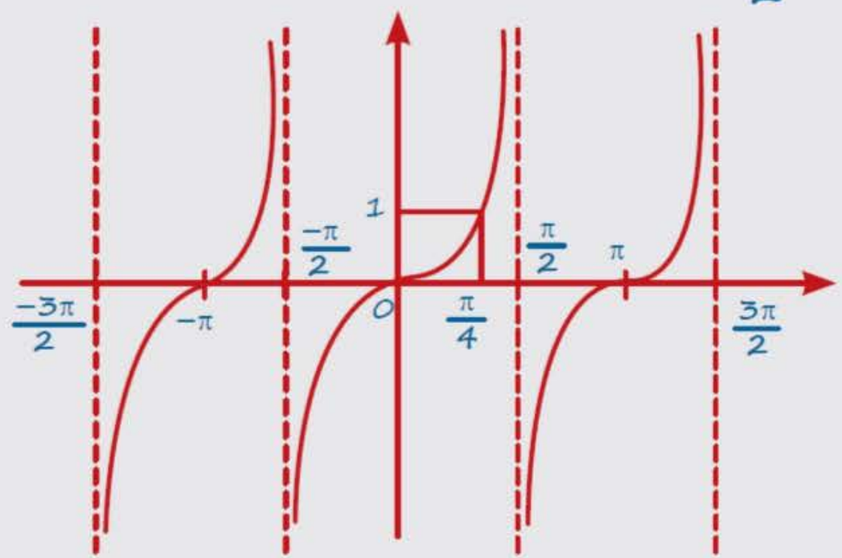
Range (cosx) ∈ [−1, 1] ✓

Period = 2π ✓

LAST CLASS RECAP

Graph of Trigonometric Functions:

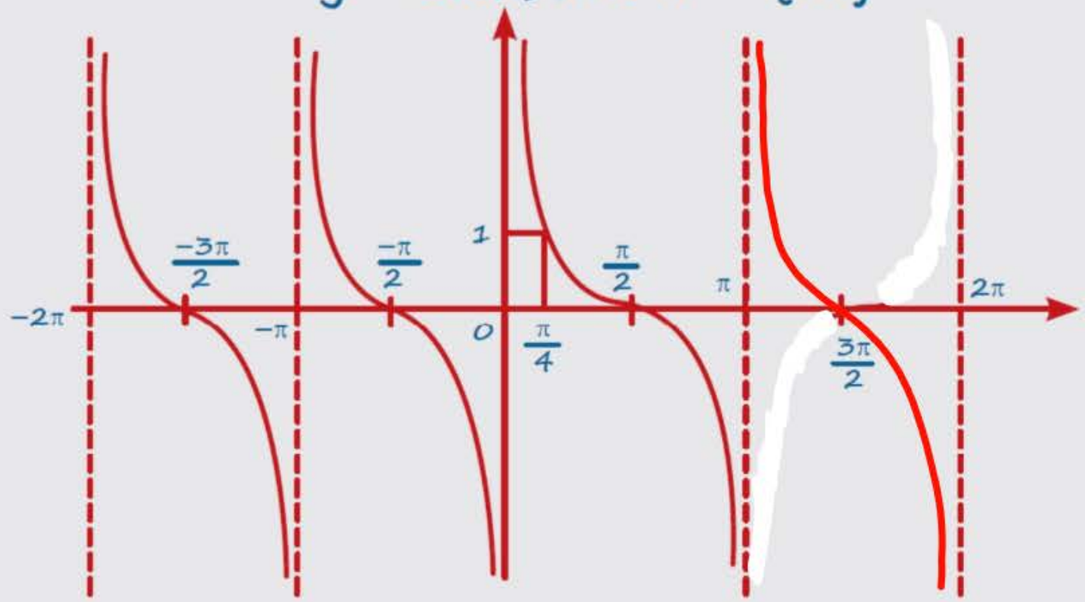
$y = \tan x,\ x \in R - \{2n+1\}\frac{\pi}{2}$



Range $\epsilon\ (-\infty, \infty) \equiv R$

Period $= \pi$

$y = \cot x,\ x \in R - \{n\pi\}$

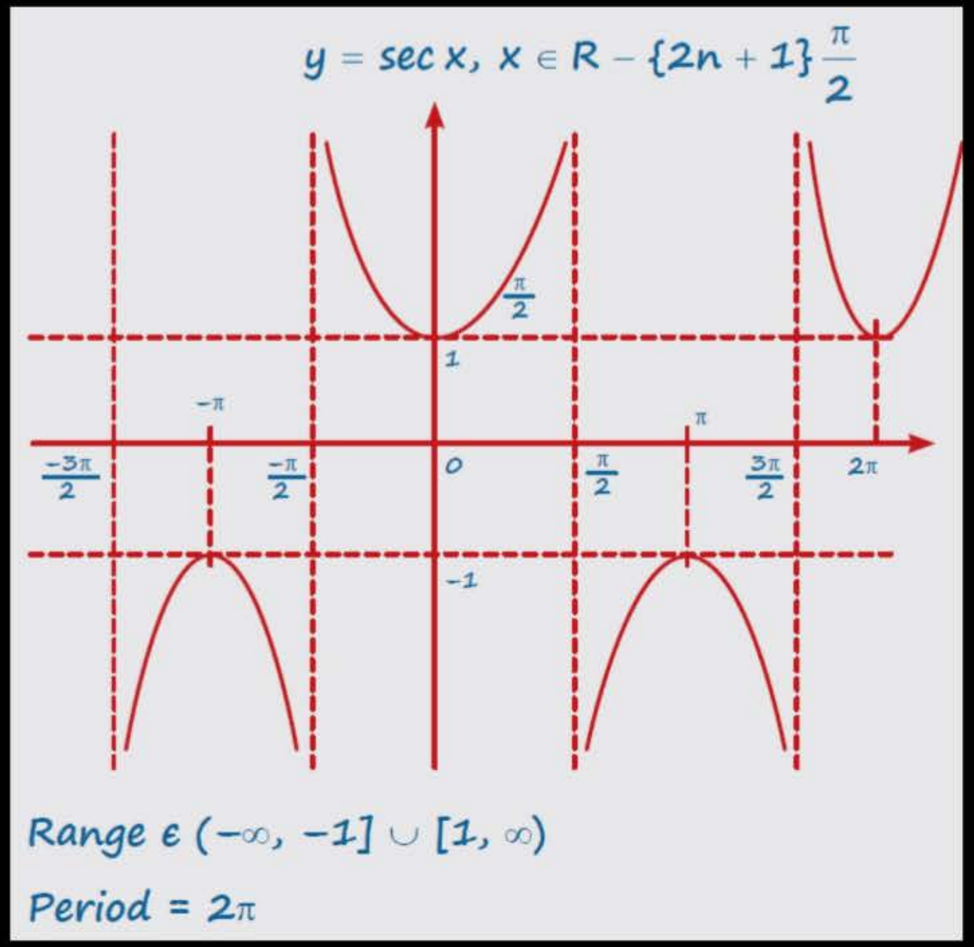


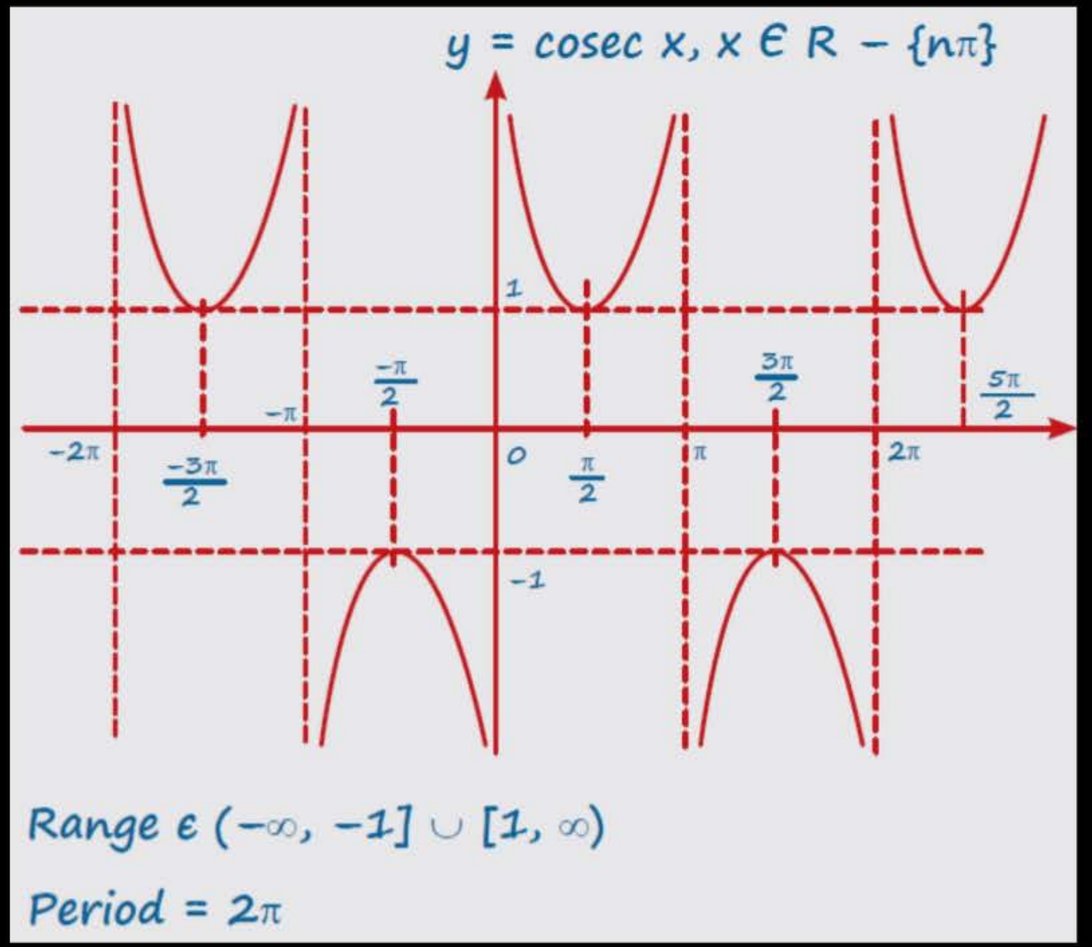
Range $\epsilon\ (-\infty, \infty) \equiv R$

Period $= \pi$

LAST CLASS RECAP

Graph of Trigonometric Functions:





SACHIN SIR SPECIAL SUMMARY(S^4)

T - Ratio	Domain	Range	Period
sinx	R	$[-1, 1]$	2π
cosx	R	$[-1, 1]$	2π
tanx	$R - (2n + 1)\pi/2$	R	π
cotx	$R - n\pi$	R	π
secx	$R - (2n + 1)\pi/2$	$(-\infty, -1] \cup [1, \infty)$	2π
cosecx	$R - n\pi$	$(-\infty, -1] \cup [1, \infty)$	2π

RAPID FIRE ROUND

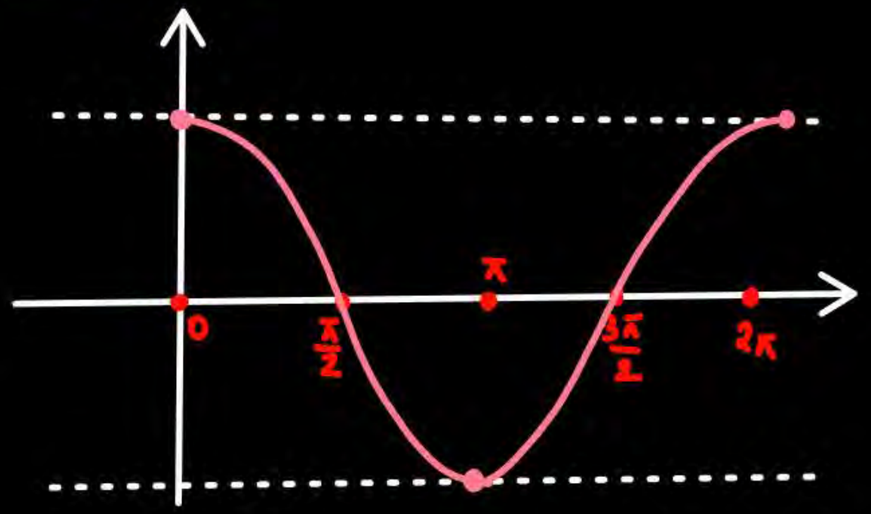
Application of graph:

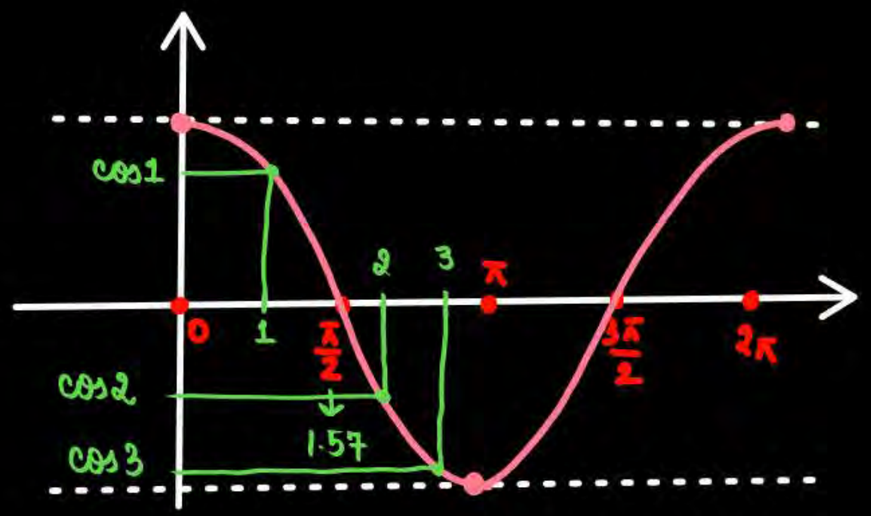
(i) For $x \in [0, 2\pi]$: $\cos x$ increases in $x \in [\pi, 2\pi]$.

(ii) For $x \in [0, 2\pi]$: $\cos x$ decreases in $x \in [0, \pi]$.

(iii) Arrange in increasing order : $\cos 1, \cos 2, \cos 3$.

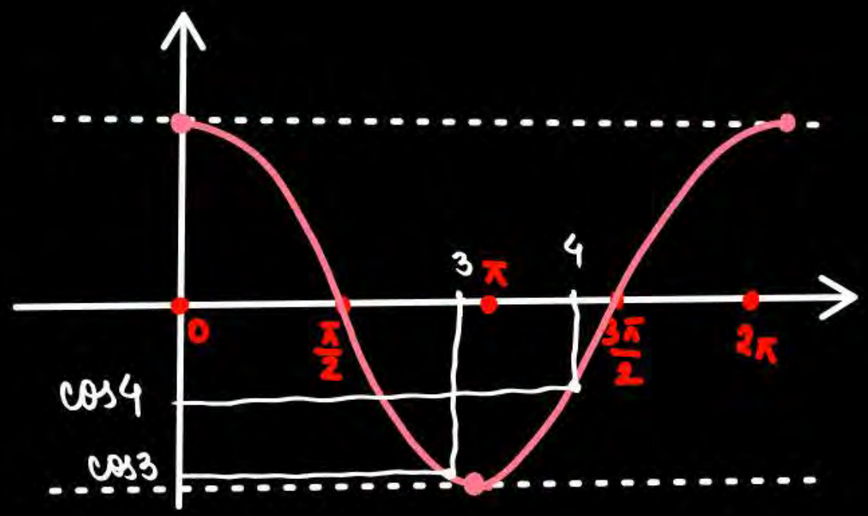
(iv) Compare: $\cos 3$ ___ $\cos 4$





(iii)

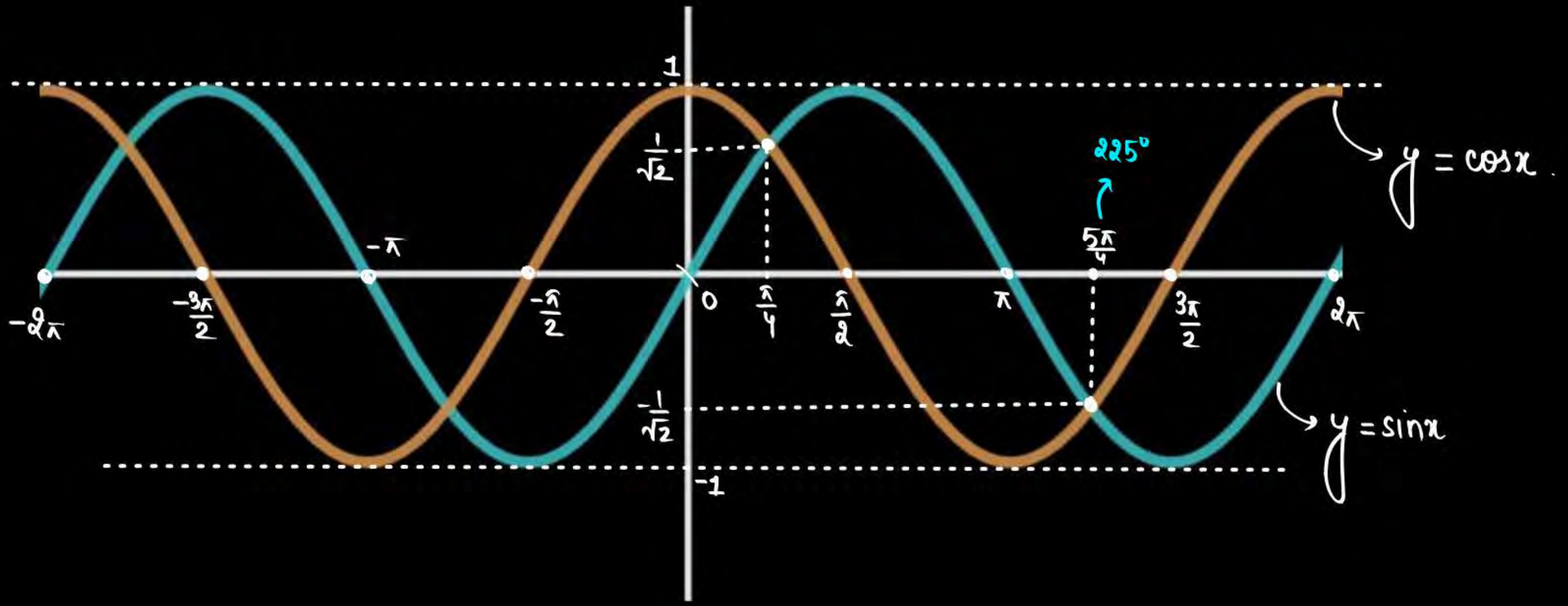
$\cos 3 < \cos 2 < \cos 1$.



(iv)

$\cos 3 \leq \cos 4$.

GRAPH OF SINE & COSINE ON SAME AXIS



RAPID FIRE ROUND

(i) If $x \in (0, \pi)$: $\cos x > \sin x$ then $x \in \left(0, \frac{\pi}{4}\right)$.

(ii) If $x \in (\pi, 2\pi)$: $\sin x > \cos x$ then $x \in \left(\pi, \frac{5\pi}{4}\right)$.

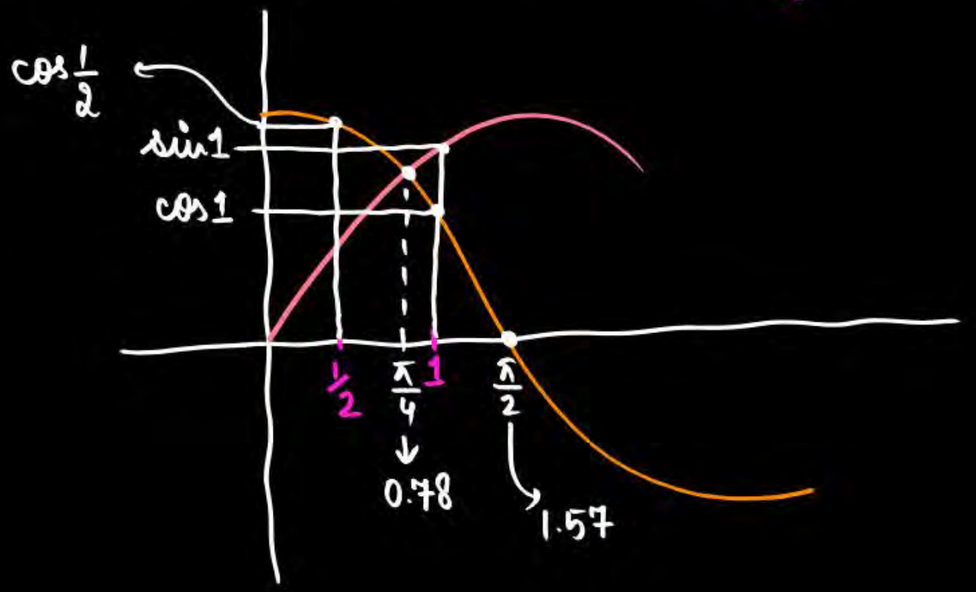
(iii) Arrange in increasing order: $\sin 1, \cos 1, \cos \frac{1}{2}$.

$\cos 1 < \sin 1 < \cos \frac{1}{2}$

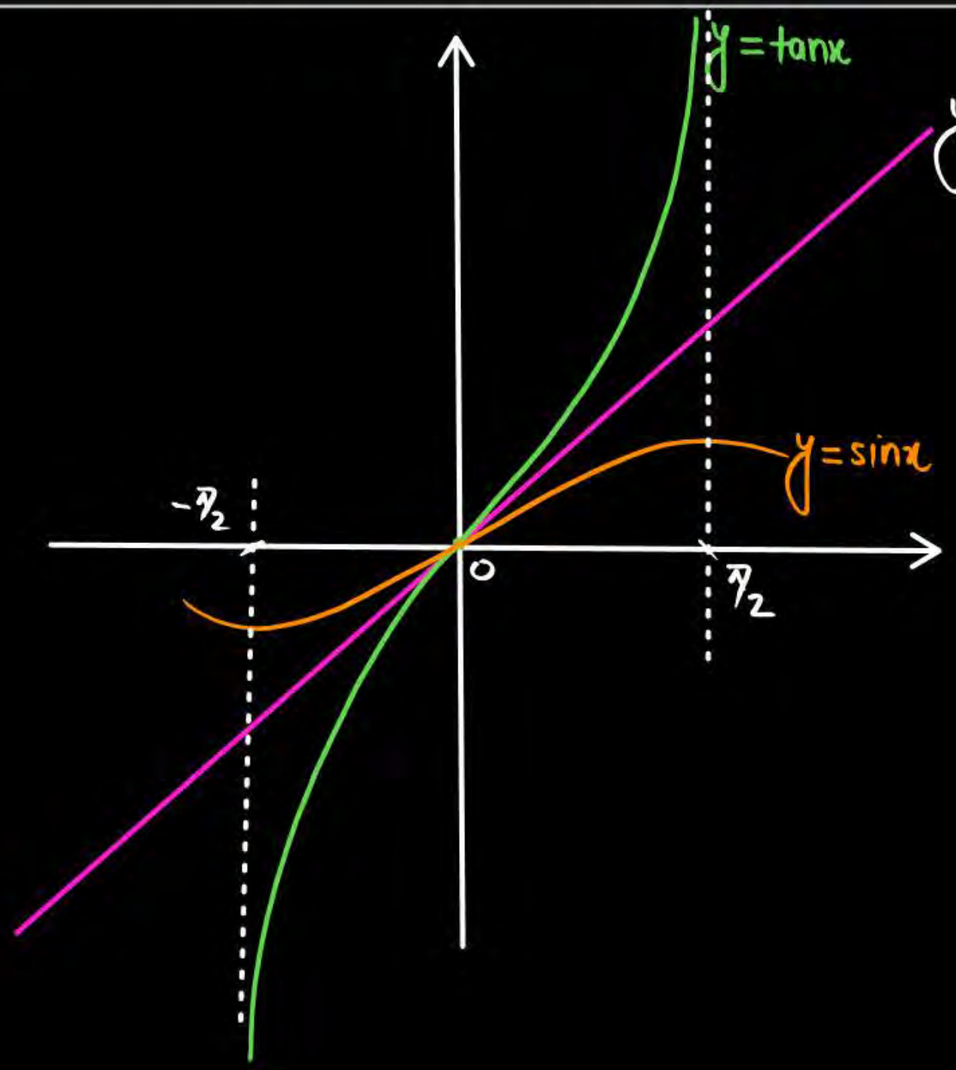
(iv) Compare: $\sin(200°) \geq \cos(200°)$

(v) Compare: $\sin(301°) ___ \cos(301°)$

∴ QiBY.



GRAPH OF SINE & TANGENT ON SAME AXIS

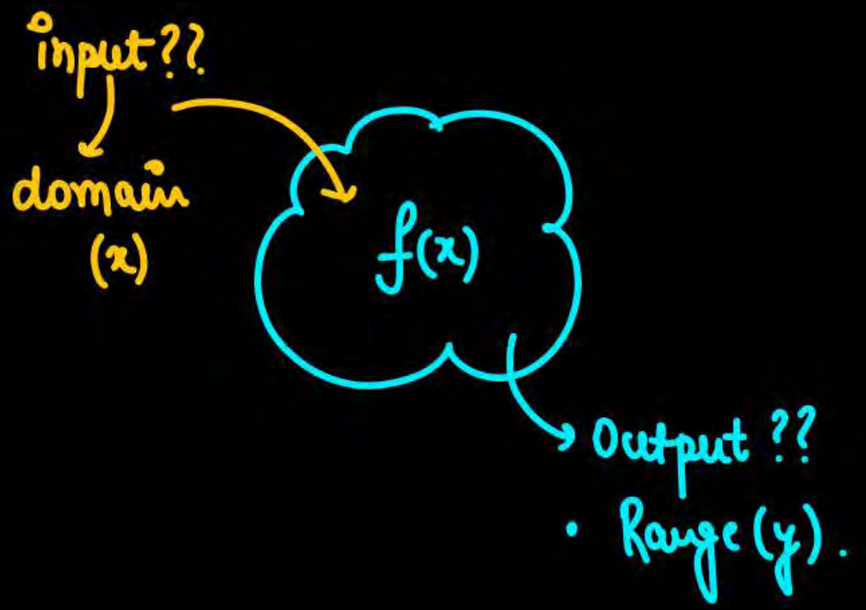


$x \in \left(0, \frac{\pi}{2}\right) \Rightarrow \sin x < x < \tan x$

$x \in \left(-\frac{\pi}{2}, 0\right) \Rightarrow \sin x > x > \tan x.$

Range.

→ Calculate
↳ "4 Types"





RANGE OF TRIGONOMETRIC FUNCTIONS

1. Find domain without simplification of given function.
2. Rules of Domain:
   (a) Denominator $\neq 0$
   (b) $\sqrt{f(x)} \Rightarrow f(x) \geq 0$
3. Always simplify given function before finding Range.



ELEMENTARY METHOD FOR FINDING RANGE

Will be applicable in Functions (Class-XII) also.

Agar koi interval given (or known) ho to, uss interval mein:

1. Addition, Subtraction, Multiplication & Division by non-zero Constant.
2. Reciprocal of an interval.
3. Squaring of an interval.

Ques:

$A \in [-3, 5)$

OR

$-3 \leq A < 5$

$\times 3$

$3A \in [-9, 15)$

$2\times$

$-6 \leq 2A < 10$

$\times \frac{1}{3}$

$-1 \leq \frac{A}{3} < \frac{5}{3}$

$\times (-6)$

$18 \geq -6A > -30$

OR

$-30 < -6A \leq 18$

(i) ~~$2A \in [-6, 10)$~~

(ii) ~~$A + 7 \in [4, 12)$~~

(iii) ~~$\frac{A}{3} \in [-1, \frac{5}{3})$~~

(iv) ~~$-6A \in (-30, 18]$~~

(v) ~~$\frac{3A+5}{2} \in [-2, 10)$~~

$3A + 5 \in [-4, 20)$

$\frac{3A+5}{2} \in [-2, 10)$

SQUARING

Example

Find y^2 in each case:

(i) $y \in [2, 5] \Rightarrow y^2 \in [4, 25]$

(ii) $y \in (-7, -3] \Rightarrow y^2 \in [9, 49)$

** (iii) $y \in (-3, 4] \Rightarrow y^2 \in [0, 16]$

** (iv) $y \in (-3, 2] \Rightarrow y^2 \in [0, 9)$

(v) $y \in [-5, 0) \Rightarrow y^2 \in (0, 25]$

(vi) $y \in [-7, -3] \Rightarrow y^2 \in [9, 49]$

(vii) $y \in [0, 3) \Rightarrow y^2 \in [0, 9)$

CUBE

Example

Find y^3 in each case:

(i) $y \in [2, 3) \Rightarrow y^3 \in [8, 27)$

(ii) $y \in (-3, -1] \Rightarrow y^3 \in (-27, -1]$

(iii) $y \in [-3, 2] \Rightarrow y^3 \in [-27, 8]$.

(1) $y \in [\alpha, \beta) \Rightarrow \frac{1}{y} \in \left(\frac{1}{\beta}, \frac{1}{\alpha}\right]$

- same sign

(2) $y \in [\alpha, \beta) \Rightarrow \frac{1}{y} \in (-\infty, \frac{1}{\alpha}] \cup \left(\frac{1}{\beta}, \infty\right)$

$\alpha < 0 < \beta > 0$.

OR

Zero interval

$\vec{H} \cdot \vec{E}$

RECIPROCAL (Notes)

Example

Find $\frac{1}{y}$ in each case:

(i) $y \in [2, 7) \Rightarrow \frac{1}{y} \in \left(\frac{1}{7}, \frac{1}{2}\right]$

(ii) $y \in (-6, -2) \Rightarrow \frac{1}{y} \in \left(-\frac{1}{2}, -\frac{1}{6}\right)$

(iii) $y \in [-2, 7) \Rightarrow \frac{1}{y} \in \left(-\infty, -\frac{1}{2}\right] \cup \left(\frac{1}{7}, \infty\right)$

(iv) $y \in [-3, 8] \Rightarrow \frac{1}{y} \in \left(-\infty, -\frac{1}{3}\right] \cup \left[\frac{1}{8}, \infty\right)$

(v) $y \in \left(-\frac{1}{2}, 3\right] \Rightarrow \frac{1}{y} \in \left(-\infty, -2\right) \cup \left[\frac{1}{3}, \infty\right)$

$y \in [-2, 7)$

$$\frac{1}{y} \in \left(-\infty, -\frac{1}{2}\right] \cup \left(\frac{1}{7}, \infty\right)$$

$y \in [-2, 0] \cup [0, 7)$

$$\frac{1}{y} \in \left(-\infty, -\frac{1}{2}\right] \cup \left(\frac{1}{7}, \infty\right)$$

$0^- = -0.000\ldots1$

$\frac{1}{0^-} = \frac{1}{-0.000\ldots1} = -\infty$

$\frac{1}{0^+} = \frac{1}{0.000\ldots1} = +\infty$

RANGE OF TRIGONOMETRIC FUNCTIONS

Type-01: Based on Range of Standard Trigonometric Functions

$\sin x \in [-1, 1]$

$\cos x \in [-1, 1]$

$\tan x \in R$

$\cot x \in R$

$\sec x \in (-\infty, -1] \cup [1, \infty)$

$\operatorname{cosec} x \in (-\infty, -1] \cup [1, \infty)$

RAR

Ques:- Range.

① $y = 3\sin x - 4$.

Sumit:- $y_{max.} = 3\times 1 - 4 = -1$.
$y_{min.} = 3(-1) - 4 = -7$

Range $\in [-7, -1]$

Santosh:-
$\sin x \in [-1, 1]$
$\times 3$ → $3\sin x \in [-3, 3]$
-4 → $3\sin x - 4 \in [-7, -1]$

② $y = \dfrac{7 - 2\cos x}{3}$.

Sumit:-
$y_{min.} = \dfrac{7 - 2\times 1}{3} = \dfrac{5}{3}$

$y_{max.} = \dfrac{7 - 2(-1)}{3} = \dfrac{9}{3} = 3$

Santosh:-
$\cos x \in [-1, 1]$
$\times(-2)$ → $-2\cos x \in [-2, 2]$
$+7$ → $7 - 2\cos x \in [5, 9]$
$\times \frac{1}{3}$ → $\dfrac{7 - 2\cos x}{3} \in \left[\dfrac{5}{3}, 3\right]$ ✓

③ $y = \dfrac{11}{5 + \cos x}$.

④ $y = \dfrac{9}{5 - 8\sin x}$

Ques:- Range.

~~(3)~~ $y = \frac{11}{5+\cos x}$.

Sumit:

$y_{max.} = \frac{11}{5+(-1)} = \frac{11}{4}$ ✓

$y_{min.} = \frac{11}{5+1} = \frac{11}{6}$ ✓

Santosh:

$\cos x \in [-1, 1]$

$5+\cos x \in [4, 6]$

$\frac{1}{5+\cos x} \in \left[\frac{1}{6}, \frac{1}{4}\right]$

$\frac{11}{5+\cos x} \in \left[\frac{11}{6}, \frac{11}{4}\right]$ ✓

(4) $y = \frac{9}{5-8\sin x}$

Sumit:

$y_{max.} = \frac{9}{5-8(1)} = \frac{9}{-3}$

$= -3$.

$y_{min.} = \frac{9}{5-8(-1)} = \frac{9}{13}$.

Santosh

$\sin x \in [-1, 1]$

$-8\sin x \in [-8, 8]$

$5-8\sin x \in [-3, 13]$

$\frac{1}{5-8\sin x} \in \left(-\infty, -\frac{1}{3}\right] \cup \left[\frac{1}{13}, \infty\right)$

$\frac{9}{5-8\sin x} \in (-\infty, -3] \cup \left[\frac{9}{13}, \infty\right)$

Question

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Find range of following functions :

(i) $y = 3\sin x - 5$ $\longrightarrow$ QIBY. – (A)

(ii) $y = \frac{1}{5\cos x + 2}$ $\longrightarrow$ QIBY – (B)

(iii) $y = \sqrt{2\sec x - 1}$

domain

$2\sec x - 1 \geq 0$

$\sec x \geq \frac{1}{2}$

$\sec x \in (-\infty, -1] \cup [1, \infty)$

$2\sec x \in (-\infty, -2] \cup [2, \infty)$

$2\sec x - 1 \in \underbrace{(-\infty, -3]}_{-ve} \cup \underbrace{[1, \infty)}_{+ve}$

$\sqrt{2\sec x - 1} \in \underbrace{[1, \infty)}_{\text{Range.}}$

Note :-

$A \in [3, 16) \rightarrow \sqrt{A} \in [\sqrt{3}, 4)$

$A \in [-5, 15) \rightarrow \sqrt{A} \in [0, \sqrt{15})$

$[-5, 15) \rightarrow [0, 15)$

Question

Page No.48

Find range of following funetions :

(iv) $y = \sin^4 x + \cos^4 x$

(v) $y = 7\sin^2 x + 2\cos^2 x + 3$

(vi) $y = \dfrac{\sin 3x}{\sin x}$

new

Sumit :-

$\sin x \in [-1, 1]$

$\sin^2 x \in [0, 1]$

$\sin^4 x \in [0, 1]$

$\cos^4 x \in [0, 1]$

add.

$\sin^4 x + \cos^4 x \in [0, 2]$

Wrong !!

Santosh :-

$y = 1 - 2\sin^2 x \cdot \cos^2 x$

$= 1 - 2(\sin x \cdot \cos x)^2$

$= 1 - 2\left(\dfrac{\sin 2x}{2}\right)^2$

$y = 1 - \dfrac{1}{2}\sin^2 2x$

Santosh:

$$y = 1 - 2\sin^2 x \cdot \cos^2 x.$$

$$= 1 - 2\left(\sin x \cdot \cos x\right)^2.$$

$$= 1 - 2\left(\frac{\sin 2x}{2}\right)^2$$

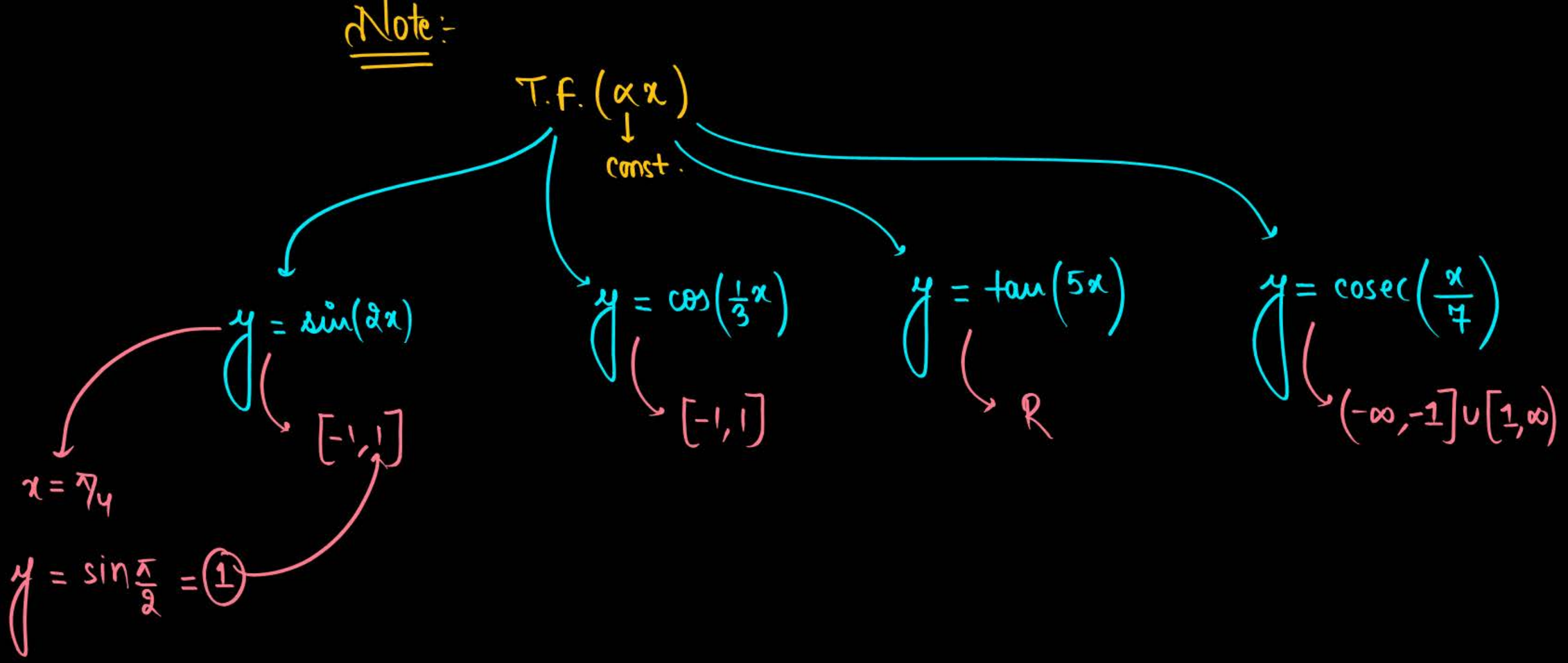
$$y = 1 - \frac{1}{2}\sin^2 2x.$$

$\sin(2x) \in [-1, 1]$

$\sin^2(2x) \in [0, 1]$

$-\frac{1}{2}\sin^2 2x \in \left[-\frac{1}{2}, 0\right]$

$1 - \frac{1}{2}\sin^2 2x \in \left[\frac{1}{2}, 1\right]$ Ans.


Note :-
T.F. (αx)
↓
const.
y = sin(2x)
[-1, 1]
x = π/4
y = sin π/2 = 1
y = cos(1/3 x)
[-1, 1]
y = tan(5x)
R
y = cosec(x/7)
(-∞, -1] ∪ [1, ∞)

RAPID FIRE DIBY'S

DIBY-A

The expression $\frac{\cos^3\alpha-\cos 3\alpha}{\cos\alpha}+\frac{\sin^3\alpha+\sin 3\alpha}{\sin\alpha}$ is equal to:

(A) 1 (B) 2 (C) 3 (D) 4

[Ans. : C]

DIBY-B

Express $\cos 5A$ in terms of $\cos A$

(A) $16\cos^5 A-20\cos^3 A+5\cos A$ (B) $16\cos^5 A-12\cos^3 A+3\cos A$

(C) $16\cos^5 A-20\cos^3 A+3\cos A$ (D) $32\cos^2 A-32\cos^3 A+6\cos A$

[Ans. : A]

DIBY-C

The value of the product $\cos 36^\circ.\cos 72^\circ$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{\sqrt{5}}{4}$

[Ans. : B]

RAPID FIRE DIBY'S

DIBY-D

Show that: (i) $\sin 105^\circ + \cos 105^\circ = \cos 45^\circ$

(ii) $\sin 75^\circ - \sin 15^\circ = \cos 105^\circ + \cos 15^\circ$

(iii) $\sin((n+1)A)\sin((n-1)A) + \cos((n+1)A)\cos((n-1)A) = \cos 2A$

DIBY-E

Let $S = \sin\sqrt{2} - \sin\sqrt{3}$ and $C = \cos\sqrt{2} - \cos\sqrt{3}$ then which one is correct?

(A) $S > 0$ and $C > 0$

(B) $S > 0$ and $C < 0$

(C) $S < 0$ and $C > 0$

(D) $S < 0$ and $C < 0$

[Ans. : A]

DIBY-F

The expression $\frac{\sec 8A - 1}{\sec 4A - 1}$ is equivalent to :

(A) $\frac{\tan 8A}{\tan 4A}$

(B) $\frac{\tan 8A}{\tan 2A}$

(C) $\frac{\tan 4A}{\tan -2A}$

(D) $\frac{\tan 2A}{\tan -8A}$

[Ans. : B]

RAPID FIRE DIBY'S

DIBY-G

What is the exact value of $\cos^2 48^\circ - \sin^2 12^\circ$

(A) $\frac{\sqrt{5}-1}{8}$ (B) $\frac{\sqrt{5}+1}{8}$ (C) $\frac{\sqrt{5}+1}{4}$ (D) $\frac{\sqrt{5}-1}{4}$

[Ans. : B]

DIBY-H

What is the exact value of $4\cos\frac{\pi}{10} - 3\sec\frac{\pi}{10} - 2\tan\frac{\pi}{10}$

(A) 0 (B) 1 (C) $\sqrt{5}$ (D) $\frac{1}{2}$

[Ans. : A]

DIBY-I

The value of the expression $\frac{\sin 50^\circ + \sin 100^\circ + \sin 210^\circ}{\sin 25^\circ \sin 50^\circ \sin 105^\circ}$ is equal to:

(A) 1 (B) 2 (C) 3 (D) 4

[Ans. : D]

RAPID FIRE DIBY'S

DIBY-J

The value of the product $\sin\frac{\pi}{16}\cdot\sin\frac{3\pi}{16}\cdot\sin\frac{5\pi}{16}\cdot\sin\frac{7\pi}{16}$ is equal to:

(A) $\frac{1}{16}$ (B) $\frac{1}{16\sqrt{2}}$ (C) $\frac{1}{8\sqrt{2}}$ (D) $\frac{1}{8}$

[Ans. : C]

DIBY-K

Find the value of $\prod_{r=1}^{7}\sin(2r-1)\frac{\pi}{14}$

(A) $\frac{1}{64}$ (B) $\frac{1}{128}$ (C) $\frac{1}{256}$ (D) $\frac{1}{32}$

[Ans. : A]

DIBY-L

If $\sin\theta+\sin\phi=a$ and $\cos\theta+\cos\phi=b$, then the value of $\tan\left(\frac{\theta-\phi}{2}\right)$ is:

(A) $\pm\sqrt{\frac{4-a^2-b^2}{a^2+b^2}}$ (B) $\pm\sqrt{\frac{a^2+b^2}{4-a^2-b^2}}$ (C) $\pm\sqrt{\frac{4+a^2+b^2}{a^2+b^2}}$ (D) $\pm\sqrt{\frac{a^2+b^2}{4+a^2+b^2}}$

[Ans. : A]

RAPID FIRE DIBY'S

DIBY-M

Show that: $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ = 0.$

DIBY-N

Show that: $\frac{\cos(A+B+C)+\cos(-A+B+C)+\cos(A-B+C)+\cos(A+B-C)}{\sin(A+B+C)+\sin(-A+B+C)-\sin(A-B+C)+\sin(A+B-C)} = \cot B$

DIBY-O

If $x - y = \frac{\pi}{4}$ and $\cot x + \cot y = 2$. then the smallest positive angles x and y are given by the ordered pair $(x, y) =$

(A) $\left(\frac{7\pi}{12}, \frac{\pi}{3}\right)$ (B) $\left(\frac{5\pi}{12}, \frac{\pi}{6}\right)$ (C) $\left(\frac{\pi}{3}, \frac{\pi}{12}\right)$ (D) $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$

[Ans. : B]

RAPID FIRE DIBY'S

DIBY-P

If α, β and γ are the interior angles of a triangle, then the value of the expression $\frac{\cos\alpha}{\sin\beta\sin\gamma} + \frac{\cos\beta}{\sin\alpha\sin\gamma} + \frac{\cos\gamma}{\sin\alpha\sin\beta}$ is equal to:

(A) 1 (B) 2 (C) 3 (D) 4

[Ans. : B]

DIBY-Q

Find the values of (1) $\sin 7\frac{1}{2}^\circ$ (2) $\cos 7\frac{1}{2}^\circ$ (3) $\tan 11\frac{1}{4}^\circ$.

[Ans. : (i) $\frac{\sqrt{4-\sqrt{2}-\sqrt{6}}}{2\sqrt{2}}$, (ii) $\frac{\sqrt{4+\sqrt{2}+\sqrt{6}}}{2\sqrt{2}}$; (iii) $-(\sqrt{2}+1)+\sqrt{4+2\sqrt{2}}$]

DIBY-R

Show that: $\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ = 1$

Bumper Question

If $A = \theta, B = \theta + \frac{\pi}{3}$ & $C = \theta - \frac{\pi}{3}$, then show that :

(i) $\tan\theta + \tan\left(\theta + \frac{\pi}{3}\right) + \tan\left(\theta - \frac{\pi}{3}\right) = 3\tan 3\theta$

(ii) $\tan\theta\tan\left(\theta + \frac{\pi}{3}\right) + \tan\theta\tan\left(\theta - \frac{\pi}{3}\right) + \tan\left(\theta + \frac{\pi}{3}\right)\tan\left(\theta - \frac{\pi}{3}\right) = -3$

(iii) $\tan\theta\tan\left(\theta + \frac{\pi}{3}\right)\tan\left(\theta - \frac{\pi}{3}\right) = -\tan 3\theta$



DO IT BY YOURSELF (DIBY)

DIBY-40

Find the exact value of $\tan^2\frac{\pi}{16}+\tan^2\frac{3\pi}{16}+\tan^2\frac{5\pi}{16}+\tan^2\frac{7\pi}{16}$

[Ans. : 28]

DIBY-41

Prove that $\cos36^\circ\cos72^\circ\cos108^\circ\cos144^\circ = 1/16$.

DIBY-42

If $A+B+C=\pi$, prove that $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2}=1-2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$.

DIBY-43

If $A+B+C=180^\circ$, prove that $\cos^2A+\cos^2B+\cos^2C=1-2\cos A\cos B\cos C$.

DIBY-44

In triangle ABC, prove that

$\sin(B+C-A)+\sin(C+A-B)+\sin(A+B-C)=4\sin A\sin B\sin C$.

DIBY-45

If $A+B+C=\pi/2$, show that

(a) $\sin^2 A+\sin^2 B+\sin^2 C=1-2\sin A\sin B\sin C$

(b) $\cos^2 A+\cos^2 B+\cos^2 C=2+2\sin A\sin B\sin C$

DIBY-46

If $\frac{\sin^4 x}{2}+\frac{\cos^4 x}{3}=\frac{1}{5}$, then [IIT-JEE-2009]

(A) $\tan^2 x=\frac{2}{3}$

(B) $\frac{\sin^8 x}{8}+\frac{\cos^8 x}{27}=\frac{1}{125}$

(C) $\tan^2 x=\frac{1}{3}$

(D) $\frac{\sin^8 x}{8}+\frac{\cos^8 x}{27}=\frac{2}{125}$

[Ans. : A, B]

DIBY-47

For $\triangle ABC$, $\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}-1=\lambda\sin\frac{\pi-A}{4}\sin\frac{\pi-B}{4}\sin\frac{\pi-C}{4}$. Then $\lambda=$

[Ans. : 4]

DIBY-48

The value of $\sin^3 10^\circ+\sin^3 50^\circ-\sin^3 70^\circ$ is equal to

(A) $-\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $-\frac{3}{4}$ (D) $-\frac{3}{8}$

[Ans. : (D)]

DIBY-49

If $\frac{\sin^4 x}{2}+\frac{\cos^4 x}{3}=\frac{1}{5}$ then the value of $\cot^2 x$ is equal to _____.

[Ans. : 3/2]

DIBY-50

The value of $\frac{\tan(\ln 6)-\tan(\ln 2)-\tan(\ln 3)}{\tan(\ln 6)\cdot\tan(\ln 2)\cdot\tan(\ln 3)}$ is equal to _____.

[Ans. : 1]

DIBY-51

The value of $\cos^3\left(\frac{\pi}{8}\right)\cdot\cos\left(\frac{3\pi}{8}\right)+\sin^3\left(\frac{\pi}{8}\right)\cdot\sin\left(\frac{3\pi}{8}\right)$ is [JEE Mains-2020]

(A) $\frac{1}{4}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

[Ans. : (B)]

DIBY-52

$\tan^6\frac{\pi}{9}-33\tan^4\frac{\pi}{9}+27\tan^2\frac{\pi}{9}$ is equal to

[Ans. : 3]

DIBY-53

If $5\left(\tan^2x-\cos^2x\right)=2\cos2x+9$, then the value of $\cos4x$ is [JEE Mains-2017]

(A) $\frac{1}{3}$ (B) $\frac{2}{9}$ (C) $-\frac{7}{9}$ (D) $-\frac{3}{5}$

[Ans. : (C)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians